Linqi Zhe

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2930 Chestnut St, Apt 1408, Philadelphia, PA 19104

Education

University of Pennsylvania — Philadelphia, PA
M.S.E. in Mechanical Engineering and Applied Mechanics; GPA: 3.80/4.00 — *Sept. 2025 – May 2027, expected*
- **Relevant Coursework:** Machine Learning, Mathematics in Robotics, Introduction to Robotics, Advanced Robotics, Foundations of Engineering Mathematics
- **Advisors:** Prof. M. Ani Hsieh and Prof. Dinesh Jayaraman, GRASP Laboratory

Zhejiang University — Hangzhou, China
B.Eng. in Chemical Engineering; GPA: 3.26/4.00 — *Sept. 2021 – May 2025*

Publications

FORGE: Fine-Tuning Over Reward Generations through Evolution — Under review, NeurIPS 2026
***Linqi Zhe**, sole author*
- An automated reward-design method that co-evolves LLM-generated reward functions with policy networks through island-based evolution and adapts policies to new rewards zero-shot through pretrained successor features. **New state of the art among fully automated LLM reward-search methods** on three MuJoCo continuous-control benchmarks.

Research Experience

FORGE: Fine-Tuning Over Reward Generations through Evolution — University of Pennsylvania
Lead Researcher | github.com/linqizhe07/zeroshotRevolve — *Oct. 2025 – May 2026*
- Developed FORGE, an automated reward-design framework that co-evolves LLM-generated reward functions and policies through island-based evolution; each child policy inherits its parent's weights and adapts to a new reward by closed-form successor-feature projection, enabling zero-shot reward adaptation without gradient-based retraining.
- **Established a new state of the art among fully automated LLM reward-search methods**: surpassed the prior best on Humanoid Locomotion and Standup at one-tenth the per-individual compute, and reached 89.1% of the published human-in-the-loop state of the art on AdroitHand door-opening from a single set of 100 expert demonstrations. Showed that the ordered reward trajectory, rather than any single reward, yields the strongest policy.
- **Technologies:** PyTorch, MuJoCo, Gymnasium, Stable-Baselines3, Hydra, OpenAI API, Weights & Biases.

Learned Dispatch Policies for Multi-Line Task Scheduling — University of Pennsylvania
Course Project, Introduction to Robotics, MEAM 5200 | Code on GitHub — *Jan. 2026 – May 2026*
- Implemented a supervised MLP, an online LLM agent, and a reinforcement-learning agent as interchangeable dispatch policies for a multi-robot assembly pipeline, selected through a single-parameter plugin interface; for the RL agent, built a custom simulator, a six-term reward function, and a from-scratch PPO implementation.
- Established a rigorous negative result: a four-symmetry analysis grounded in classical queueing theory showed that no learned policy can outperform First-Come-First-Served under the problem's symmetries, and identified the symmetry-breaking conditions under which a learned policy could.

Autonomous Robotic Car — University of Pennsylvania
Course Project, Advanced Mechatronic System Design — *Jan. 2026 – May 2026*
- Designed and built an autonomous robotic car on ESP32 and ATmega32U4 microcontrollers, implementing low-level motor control, sensor fusion for real-time obstacle avoidance, and interrupt-driven safety handling.

Technical Skills

ML & RL: PyTorch, Stable-Baselines3, SAC, PPO, RND, successor features, HILP, evolutionary search
LLM / VLM: OpenAI GPT-4 and GPT-4V APIs, Qwen3-Coder, prompt engineering for reward generation, evolutionary prompt design
Programming: Python, C/C++, MATLAB
RL Environments & Simulation: MuJoCo, Gymnasium, Gymnasium-Robotics, D4RL
Tools & Infrastructure: Git, Linux, Hydra, Weights & Biases, TensorBoard, Docker, multi-GPU training
Embedded Systems: ESP32, ATmega32U4; PWM, timer, and interrupt programming
Languages: Fluent English, native Chinese

Awards & Honors

Second Prize, National College Student Chemical Engineering Design Competition — 2024
Third Prize, Zhejiang Province College Students' Physics Innovation Competition — 2023